# Beyond Confidence: An Analysis of Cross-View Disagreement for Admission and Readout in Test-Time CLIP Memory

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CV801: Project Teaser Video

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# Problem

## What should a test-time CLIP memory store?

### Test-Time Memory for CLIP

- CLIP is frozen. Test images arrive one at a time, without labels.
- Methods such as TDA store past test images with the labels CLIP predicted for them. Similar stored images vote on new predictions.[a]
- TDA decides what to store by confidence, but confident predictions can still be wrong.

[a] Zhang et al., *Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification*.

# Idea

## Use the disagreement between an image and its flip

### Flip Disagreement

- Run CLIP on the image and on its horizontal flip, and measure how much the two predictions differ with Jensen–Shannon (JS) divergence.
- Admission: store an image only if its JS is small.
- Readout: stored images with a large JS vote less.
- Question: does JS tell us anything that confidence does not?

# Experimental Setup

## Data, model and Process

### Setup

- Office-Home: 4 domains, 65 classes. Each domain is split per class into dev (10%), test (70%) and held-out (20%).
- CLIP ViT-B/16. Cache settings from TDA: 3 slots per class, $\alpha = 2$, $\beta = 5$.

### How the Cache Scores a Test Image

$$s_c(x) = z_c(x) + \alpha \sum_j \exp(-\beta\,(1 - u^\top k_j))\, r_j\, \mathbb{I}[y_j = c], \qquad r_j = \exp\left(-\gamma\, \frac{J_j}{\ln 2}\right)$$

$$J(x) = H\left(\frac{p^{(0)} + p^{(1)}}{2}\right) - \frac{1}{2}\left[H(p^{(0)}) + H(p^{(1)})\right]$$

- $z_c$: CLIP score for class $c$. $u$: test image feature. $k_j$, $y_j$, $J_j$: feature, predicted label and JS of stored image $j$. $p^{(0)}$, $p^{(1)}$: probabilities for the original and flipped image.
- $\alpha$: weight of the cache votes ($\alpha = 0$ is zero-shot CLIP). $\beta$: how fast a vote fades with lower similarity. $\gamma$: how much high-JS entries are down-weighted ($\gamma = 0$ is the ordinary cache).

# Conditions

## JS conditions and the controls they are compared with

### Conditions Compared

| ID | Admission | Readout |
|---|---|---|
| B0 | – (zero-shot CLIP) | – |
| C0 | all images | plain |
| P1 | low JS | plain |
| P2 | all images | JS-weighted |
| P3 | low JS | JS-weighted |
| B2 | low mean view entropy | plain |
| B3 | low entropy of the mean | plain |
| B4 | random | plain |
| B6 | class-wise entropy threshold | plain |
| BA | both views agree | plain |

Table: Admission rules are compared at the same share of admitted images (25, 50 and 75%) where possible.

# Is Disagreement Informative?

## RQ1: how much the two views differ

### Distribution of JS

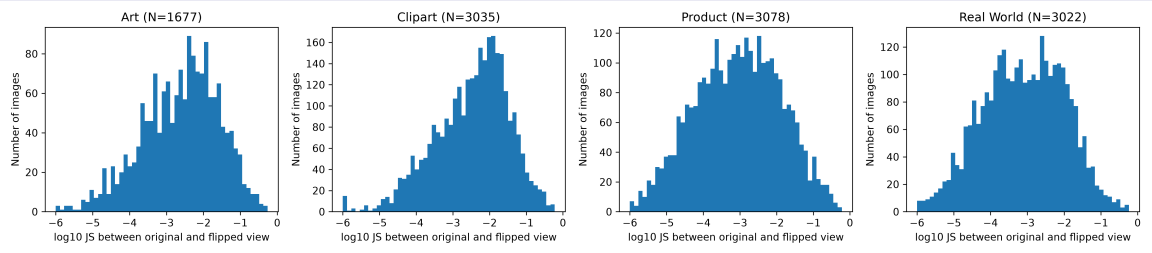



Figure: JS between the original and flipped view on the test split, on a log scale. JS is below 0.01 for 64–83% of images, depending on the domain. The bar at $-6$ collects all values at or below $10^{-6}$.

# Is Disagreement Informative?

## RQ1: accuracy of zero-shot predictions by entropy and JS

### Accuracy by Entropy Decile

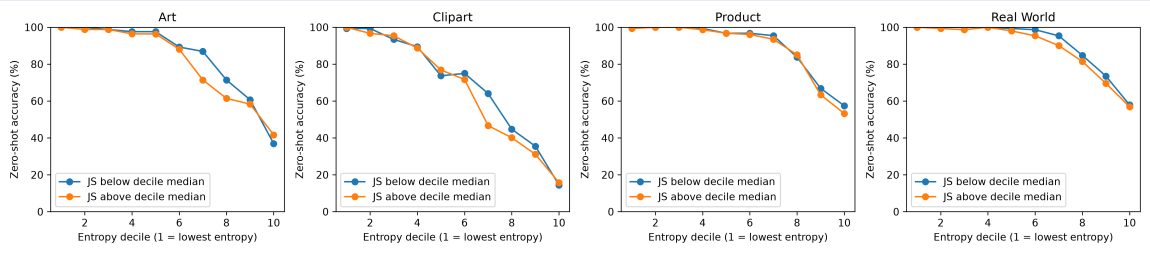



Figure: Zero-shot accuracy in each entropy decile, for images below and above the decile's median JS. The lines separate mostly in the higher-entropy deciles.

# Is Disagreement Informative?

## RQ1: separating correct from wrong predictions

### AUROC of Each Score

| Domain | Entropy | JS | JS within entropy deciles [95% CI] |
|---|---|---|---|
| Art | 0.870 | 0.788 | 0.593 [0.505, 0.645] |
| Clipart | 0.862 | 0.722 | 0.511 [0.473, 0.555] |
| Product | 0.883 | 0.795 | 0.543 [0.431, 0.598] |
| Real World | 0.883 | 0.820 | 0.644 [0.577, 0.699] |

Table: 0.5 is chance. Within entropy deciles, JS is above chance only for Real World and, barely, Art.

# Does Disagreement Help the Cache?

## RQ2 and RQ3: accuracy with 3 slots per class

### Accuracy and Difference to C0

| Condition | Accuracy (%) | vs C0 | Wins vs C0 |
|---|---|---|---|
| B0 | 82.30 | $-0.53$ | 0/12 |
| C0 | 82.83 | 0.00 | – |
| P1 | 82.07 | $-0.76$ | 0/12 |
| P2 | 82.50 | $-0.32$ | 1/12 |
| P3 | 82.06 | $-0.77$ | 0/12 |
| B2 | 81.54 | $-1.28$ | 2/12 |
| B3 | 81.52 | $-1.30$ | 2/12 |
| B4 | 82.70 | $-0.12$ | 4/12 |
| B6 | 81.92 | $-0.91$ | 0/12 |
| BA | 82.81 | $-0.02$ | 4/12 |

Table: Mean over 4 domains and 3 orders, at 50% admission and $\gamma = 16$. Wins counts the 12 domain–order pairs where a condition beats C0.

# Does Disagreement Help the Cache?

## RQ2: accuracy at different admission rates

### Accuracy against Admitted Share

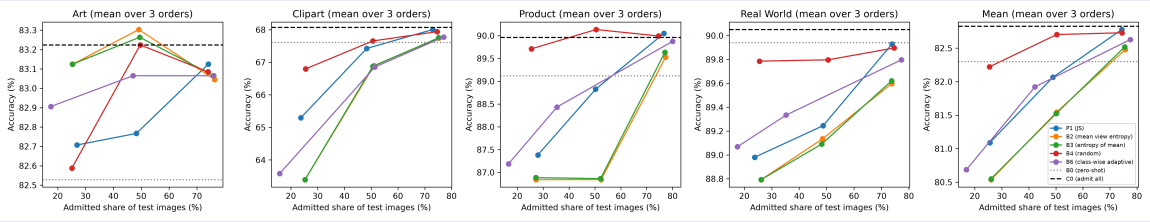



Figure: Accuracy rises as more images are admitted. On average over domains, no rule beats admitting all images (C0).

# Does Disagreement Help the Cache?

## RQ3: effect of the readout weight $\gamma$

### Accuracy against $\gamma$

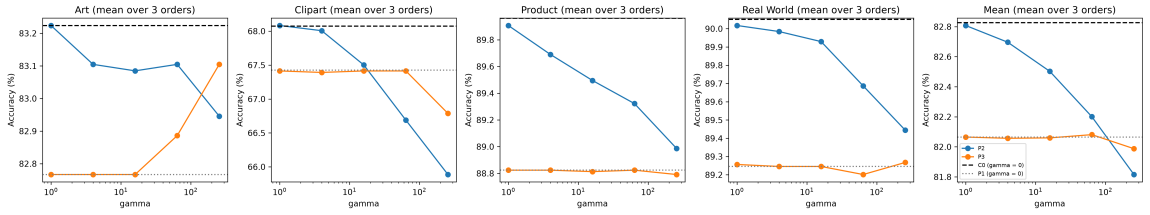



Figure: P2 weights stored images by JS. P3 also admits only low-JS images, at 50% admission. P2 loses accuracy as $\gamma$ grows, while P3 stays close to P1.

# Why Selective Admission Hurts

## Classes that CLIP is unsure about get fewer cache entries

### Unfilled Classes and Accuracy

| | 3 slots per class | | 8 slots per class | |
|---|---|---|---|---|
| Condition | Not full | vs C0 | Not full | vs C0 |
| C0 | 0.25 | 0.00 | 1.50 | 0.00 |
| B4 | 1.00 | −0.12 | 5.58 | −0.66 |
| B6 | 2.92 | −0.91 | 9.92 | −2.43 |
| P1 | 5.00 | −0.76 | 15.00 | −2.50 |
| B2 | 7.00 | −1.28 | 15.75 | −3.17 |
| B3 | 7.25 | −1.30 | 16.00 | −3.24 |

Table: Classes (of 65) whose slots are not all filled at the end of the stream. With 8 slots, the losses grow and follow the number of unfilled classes.

# Why Selective Admission Hurts

## Fewer wrong labels in the cache do not mean better predictions

### A Cleaner Cache Is Not Better

| Condition | Wrong stored labels (%) | Classes with entries | vs C0 |
|---|---|---|---|
| C0 | 3.04 | 65.00 | 0.00 |
| B4 | 4.26 | 65.00 | $-0.12$ |
| P1 | 2.77 | 62.50 | $-0.76$ |
| B2 | 2.02 | 62.50 | $-1.28$ |
| B3 | 2.03 | 62.25 | $-1.30$ |

Table: With 3 slots per class. Random admission stores the most wrong labels and stays closest to C0.

# Progress and Next Steps

## What is done and the plan for the final report

### Progress So Far

- Code for data loading, CLIP features, the online cache and the analysis is complete and tested.[a]
- Features for both views are extracted for all 4 domains. All 10 conditions are run at 3 admission rates, 5 values of $\gamma$ and 3 stream orders (456 runs per cache size), with 3 and 8 slots per class.
- Checks pass: B0 equals zero-shot CLIP, and the readout never changes what is stored.
- All tables and figures in this talk come from these runs.[b]

[a] Code: https://github.com/soumenkm/MBZUAI_CV801_Final_Project/tree/main/src

[b] Results: https://github.com/soumenkm/MBZUAI_CV801_Final_Project/tree/main/outputs

### Next Steps

- Average votes within each class instead of summing them, to test the explanation directly.
- Add a readout control that keeps the total vote mass fixed.
- Use JS for medium-entropy images, where it carries some information.
- Measure how much each stored entry helps, on the held-out split.

## Thank You

Any Questions? Suggestions?